

Database Technologies-Assignment 1

1. Create 'spj' database'. (Database for suppliers, parts and jobs.
2. use spj database.

```
C:\Users\91879>mysql -u KD2_98846_Raj -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 24
Server version: 8.4.6 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

KD2_98846_Raj>SHOW TABLES;
ERROR 1046 (3D000): No database selected
KD2_98846_Raj>Show database
-> Show database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
syntax to use near 'database' at line 1
Show database' at line 1
KD2_98846_Raj>Show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| spj |
| sys |
+-----+
5 rows in set (0.04 sec)

KD2_98846_Raj>use spj database;
Database changed
```

3. Create following tables in spj database.

CREATE TABLE S (S# CHAR(5) NOT NULL, Sname CHAR(20), status SMALLINT, city CHAR(15));

CREATE TABLE P (P# CHAR(6) NOT NULL, Pname CHAR(20), color CHAR(6), weight SMALLINT, city CHAR(15));

CREATE TABLE J (J# CHAR(4) NOT NULL, Jname CHAR(10), City CHAR(15));

CREATE TABLE sp (S# CHAR(4) NOT NULL, P# CHAR(4) NOT NULL, J# CHAR(4) NOT NULL, QTY INT);

```

KD2_98846_Raj>CREATE TABLE S (`S#` CHAR(5) NOT NULL,Sname CHAR(20),status SMALLINT,city CHAR(15));
Query OK, 0 rows affected (0.10 sec)

KD2_98846_Raj>show table S;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right
syntax to use near 'S' at line 1
KD2_98846_Raj>show tables;
+-----+
| Tables_in_spj |
+-----+
| s              |
+-----+
1 row in set (0.00 sec)

KD2_98846_Raj>CREATE TABLE P (`P#` char(6) NOT NULL, Pname CHAR(20),color CHAR(6),weight SMALLINT,city CHAR(15));
Query OK, 0 rows affected (0.10 sec)

KD2_98846_Raj>CREATE TABLE J (`J#` CHAR(4) NOT NULL, Jname CHAR(10),city CHAR(15));
Query OK, 0 rows affected (0.09 sec)

KD2_98846_Raj>SHOW TABLES;
+-----+
| Tables_in_spj |
+-----+
| j              |
| p              |
| s              |
+-----+
3 rows in set (0.00 sec)

KD2_98846_Raj>CREATE TABLE sp (`S#` CHAR(4) NOT NULL, P# CHAR(4) NOT NULL,J#CHAR(4) NOT NULL,QTY INT);
-> ^C
KD2_98846_Raj>CREATE TABLE sp (`S#` CHAR(4) NOT NULL, `P#` CHAR(4) NOT NULL,J#CHAR(4) NOT NULL,QTY INT);
-> CREATE TABLE sp (`S#` CHAR(4) NOT NULL, `P#` CHAR(4) NOT NULL,`J#` CHAR(4) NOT NULL,QTY INT);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right

```

```

KD2_98846_Raj>CREATE TABLE sp (`S#` CHAR(4) NOT NULL, `P#` CHAR(4) NOT NULL,`J#` CHAR(4) NOT NULL,QTY INT);
Query OK, 0 rows affected (0.08 sec)

KD2_98846_Raj>DESCRIBE S;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S#    | char(5) | NO   |     | NULL    |       |
| Sname | char(20) | YES  |     | NULL    |       |
| status | smallint | YES  |     | NULL    |       |
| city  | char(15) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.02 sec)

KD2_98846_Raj>INSERT INTO S VALUES('S1','Smith',20,'London'),('S2','Jones',10,'Paris'),('S3','Blake',30,'Paris'),('S4','Clark',20,'Lon
don'),('S5','Adams',30,'Athens');
Query OK, 5 rows affected (0.03 sec)
Records: 5 Duplicates: 0 Warnings: 0

KD2_98846_Raj>INSERT INTO P VALUES('P1','Nut','Red',12,'London'),('P2','Bolt','Green',17,'Paris'),('P3','Screw','Blue',17,'Rome'),('P4
','Screw','Red',14,'London'),('P5','Cam','Blue',12,'Paris'),('P6','Cog','Red',19,'London');
Query OK, 6 rows affected (0.05 sec)
Records: 6 Duplicates: 0 Warnings: 0

KD2_98846_Raj>INSERT INTO J VALUES('J1','Sorter','Paris'),('J2','Punch','Rome'),('J3','Reader','Athens'),('J4','Console','Athens'),('J
5','Collector','London'),('J6','Terminal','Oslo'),('J7','Tape','London');
Query OK, 7 rows affected (0.02 sec)
Records: 7 Duplicates: 0 Warnings: 0

KD2_98846_Raj>INSERT INTO SP VALUES
-> ('S1','P1','J1',200),
-> ('S1','P1','J4',700),
-> ('S2','P3','J1',400),
-> ('S2','P3','J2',200),

```

```
5','Collector','London'),('J6','Terminal','Oslo'),('J7','Tape','London');
Query OK, 7 rows affected (0.02 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
KD2_98846_Raj>INSERT INTO SP VALUES
```

```
-> ('S1','P1','J1',200),
-> ('S1','P1','J4',700),
-> ('S2','P3','J1',400),
-> ('S2','P3','J2',200),
-> ('S2','P3','J3',200),
-> ('S2','P3','J4',500),
-> ('S2','P3','J5',600),
-> ('S2','P3','J6',400),
-> ('S2','P3','J7',800),
-> ('S2','P5','J2',100),
-> ('S3','P3','J1',200),
-> ('S3','P4','J2',500),
-> ('S4','P6','J3',300),
-> ('S4','P6','J7',300),
-> ('S5','P2','J2',200),
-> ('S5','P2','J4',100),
-> ('S5','P5','J5',500),
-> ('S5','P5','J7',100),
-> ('S5','P6','J2',200),
-> ('S5','P1','J4',100),
-> ('S5','P3','J4',200),
-> ('S5','P4','J4',800),
-> ('S5','P5','J4',400),
-> ('S5','P6','J4',500);
```

```
Query OK, 24 rows affected (0.01 sec)
Records: 24 Duplicates: 0 Warnings: 0
```

```
KD2_98846_Raj>exit;
Bye
```

```
KD2_98846_Raj>use spj database;
```

```
Database changed
```

```
KD2_98846_Raj>show tables;
```

```
+-----+
| Tables_in_spj |
+-----+
| j              |
| p              |
| s              |
| sp             |
+-----+
```

```
4 rows in set (0.00 sec)
```

```
KD2_98846_Raj>SELECT*FROM S;
```

```
+-----+-----+-----+-----+
| S# | Sname | status | city |
+-----+-----+-----+-----+
| S1 | Smith | 20     | London |
| S2 | Jones | 10     | Paris |
| S3 | Blake | 30     | Paris |
| S4 | Clark | 20     | London |
| S5 | Adams | 30     | Athens |
+-----+-----+-----+-----+
```

```
5 rows in set (0.04 sec)
```

```
KD2_98846_Raj>SELECT*FROM P;
```

```
+-----+-----+-----+-----+-----+
| P# | Pname | color | weight | city |
+-----+-----+-----+-----+-----+
| P1 | Nut   | Red   | 12     | London |
| P2 | Bolt  | Green | 17     | Paris |
| P3 | Screw | Blue  | 17     | Rome |
| P4 | Screw | Red   | 14     | London |
| P5 | Cam   | Blue  | 12     | Paris |
+-----+-----+-----+-----+-----+
```

```

6 rows in set (0.00 sec)

KD2_98846_Raj>SELECT*FROM J;
+----+-----+-----+
| J# | Jname | city |
+----+-----+-----+
| J1 | Sorter | Paris |
| J2 | Punch  | Rome  |
| J3 | Reader | Athens |
| J4 | Console | Athens |
| J5 | Collector | London |
| J6 | Terminal | Oslo |
| J7 | Tape   | London |
+----+-----+-----+
7 rows in set (0.00 sec)

KD2_98846_Raj>SELECT*FROM SP;
+----+----+----+----+
| S# | P# | J# | QTY |
+----+----+----+----+
| S1 | P1 | J1 | 200 |
| S1 | P1 | J4 | 700 |
| S2 | P3 | J1 | 400 |
| S2 | P3 | J2 | 200 |
| S2 | P3 | J3 | 200 |
| S2 | P3 | J4 | 500 |
| S2 | P3 | J5 | 600 |
| S2 | P3 | J6 | 400 |
| S2 | P3 | J7 | 800 |
| S2 | P5 | J2 | 100 |
| S3 | P3 | J1 | 200 |
| S3 | P4 | J2 | 500 |
| S4 | P6 | J3 | 300 |
| S4 | P6 | J7 | 300 |

```

Answer the following questions (by observations only – not queries).

1. How many orders data is present in the database?
ans - 10 customers
2. How many customers are present in the database?
ans- 7 customers
3. How many salespeople are present in the database?
ans- 5 salespeople
4. On which date order with highest amount is placed?
ans-1990-10-04
5. For which order salesman got maximum commission? Hint: you need to refer data of two tables.
ans-Order 3011 — Salesman Peel
6. Which salesmen are handling more than two customers? Hint: you need to refer data of two tables.
ans-No salesman is handling more than two customers.
7. Which city have single customer?
ans-Berlin
8. Which city has multiple salespeople?
ans-London
9. What is name of customer and salesman of the maximum amount order?
ans-Customer = Clemens, Salesman = Peel
10. Study relationship between the tables in sales database.
ans- relationships are mainly one to many.

